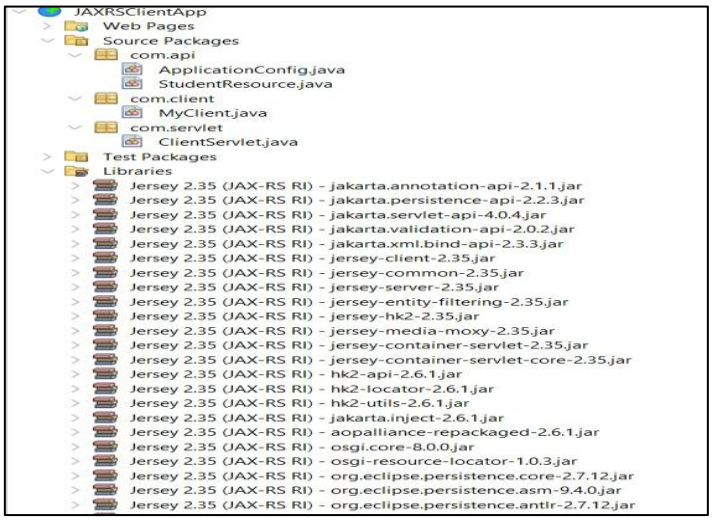
PRACTICAL 14

JAX-RS Client Develop a JAX-RS client that consumes RESTful service. Utilize the client in UI layer (JSP pages)



CODE:

index.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<html>
<head>
   <title>JAX-RS Client</title>
</head>
<body>
<h2>Student Data</h2>
<pre>
${data}
</pre>
</body>
```

```
</html>
```

ClientServlet.java

```
package com.servlet;

import com.client.MyClient;

import java.io.IOException;

import javax.servlet.*;

import javax.servlet.annotation.WebServlet;

import javax.servlet.http.*;


@WebServlet("/client")

public class ClientServlet extends HttpServlet {


    @Override

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)

            throws ServletException, IOException {


        String result = MyClient.getData();


        request.setAttribute("data", result);


        request.getRequestDispatcher("index.jsp").forward(request, response);

    }

}
```

MyApplication.java

```
package com.api;
import javax.ws.rs.ApplicationPath;
import javax.ws.rs.core.Application;

@ApplicationPath("/webapi")
public class ApplicationConfig extends Application {
}
```

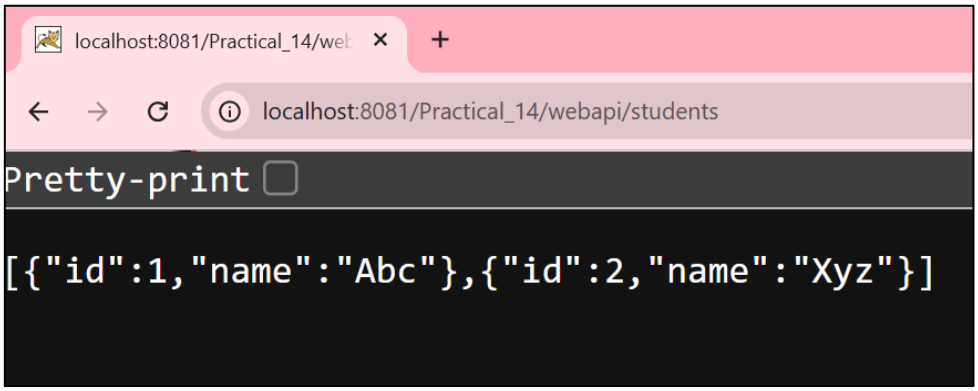
MyClient.java

```
package com.client;
import javax.ws.rs.client.*;
import javax.ws.rs.core.MediaType;
public class MyClient {

    private static final String URL =
"http://localhost:8080/Practical_14/webapi/students";

    public static String getData() {
        Client client = ClientBuilder.newClient();
        String response = client
                .target(URL)
                .request(MediaType.APPLICATION_JSON)
                .get(String.class);
        client.close();
        return response;
    }
}
```

StudentResource.java

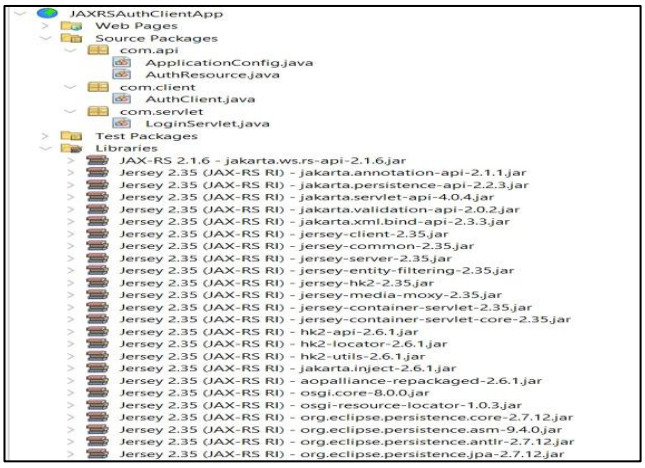
```
package com.api;

import javax.ws.rs.*;

import javax.ws.rs.core.MediaType;


@Path("/students")

public class StudentResource {


    @GET

    @Produces(MediaType.APPLICATION_JSON)

    public String getStudents() {

        return "[{\"id\":1,\"name\":\"Abc\"},{\"id\":2,\"name\":\"\"}]";

    }

}
```

OUTPUT:



PRACTICAL 15

JAX-RS Client Develop a RESTful service to perform authentication of the user. The JAX-RS client will be utillized in servlet and provide interface between jsp page and web service to perform authentication.



CODE:

index.jsp

```
<%@ page contentType="text/html;charset=UTF-8" %>
<html>
<head>
    <title>Login Page</title>
</head>
<body>
<h2>Login Form</h2>
<form action="login" method="post">
    Username: <input type="text" name="username"><br><br>
    Password: <input type="password" name="password"><br><br>
    <input type="submit" value="Login">
</form>
```

```
<br>
<h3>
   ${message}
</h3>
</body>
</html>
```

AuthClient.java

```
package com.client;
import javax.ws.rs.client.*;
import javax.ws.rs.core.*;

public class AuthClient {
   private static final String URL =
   "http://localhost:8080/Practical_15/webapi/auth";
   public static String authenticate(String username, String password) {
      System.out.println("Calling API: " + URL)
      Client client = ClientBuilder.newClient();
      Form form = new Form();
      form.param("username", username);
      form.param("password", password);

      String response = client.target(URL)
            .request(MediaType.TEXT_PLAIN)
            .post(Entity.entity(form,
MediaType.APPLICATION_FORM_URLENCODED),
                  String.class);
```

```
        client.close();
        return response;
    }
}
```

AuthResource.java

```
package com.api;
import javax.ws.rs.*;
import javax.ws.rs.core.MediaType;

@Path("/auth")
public class AuthResource {

    @POST
    @Consumes(MediaType.APPLICATION_FORM_URLENCODED)
    @Produces(MediaType.TEXT_PLAIN)
    public String login(@FormParam("username") String username,
                        @FormParam("password") String password) {
        System.out.println("API Called: " + username);
        if ("admin".equals(username) && "1234".equals(password)) {
            return "SUCCESS";
        } else {
            return "FAIL";
        }
    }
}
```

LoginServlet.java

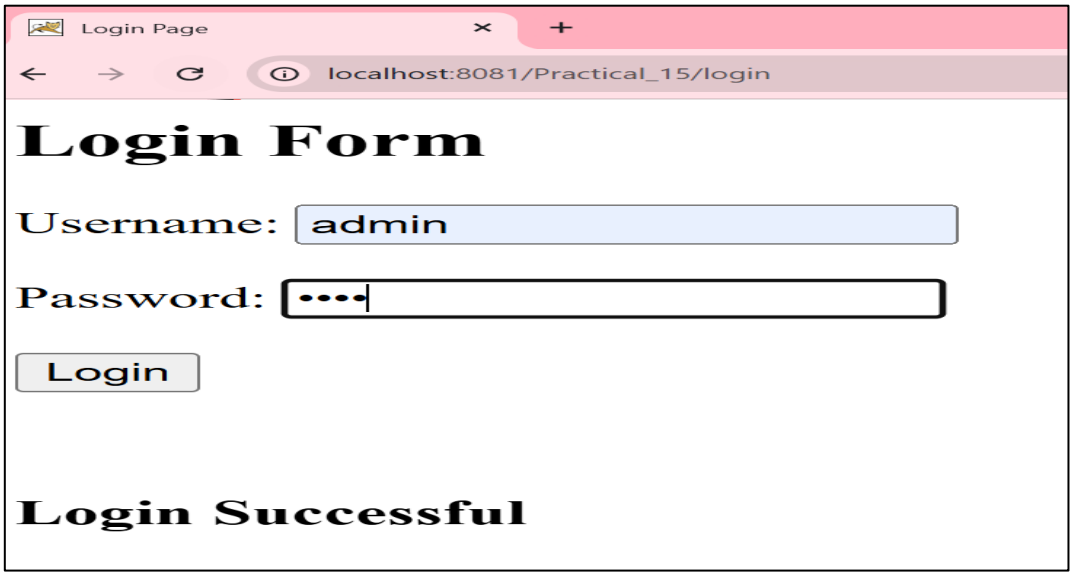
```
package com.servlet;

import com.client.AuthClient;
import java.io.IOException;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {
        String username = request.getParameter("username");
        String password = request.getParameter("password");
        System.out.println("Servlet Received: " + username);
        String result = AuthClient.authenticate(username, password);

        if ("SUCCESS".equals(result)) {
            request.setAttribute("message", "Login Successful!");
        } else {
            request.setAttribute("message", "Invalid Credentials!");
        }
```

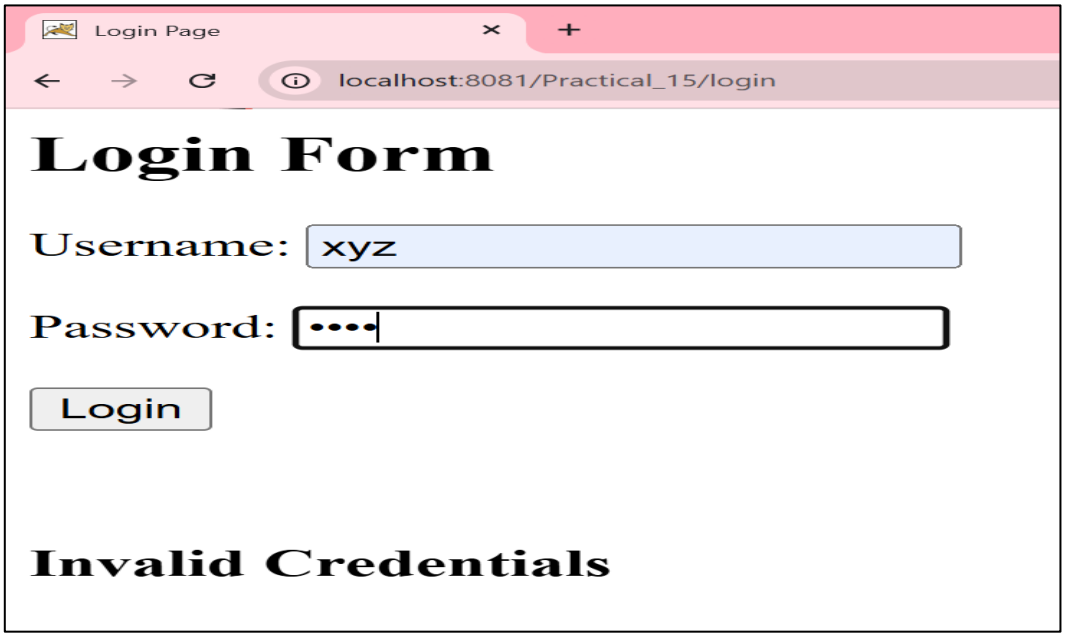
```
        request.getRequestDispatcher("index.jsp").forward(request, response);
    }
}
```

MyApplication.java

```
package com.api;
import javax.ws.rs.ApplicationPath;
import javax.ws.rs.core.Application;

@ApplicationPath("/webapi")
public class ApplicationConfig extends Application {
}
```

OUTPUT:


Login Page
localhost:8081/Practical_15/login
Login Form
Username: admin
Password: ••••
Login
Login Successful


Login Page
localhost:8081/Practical_15/login
Login Form
Username: xyz
Password: ••••
Login
Invalid Credentials